

α -tempered Variational Bayes over Sparse Variational Gaussian Process

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Motivation—Kriging Does What?

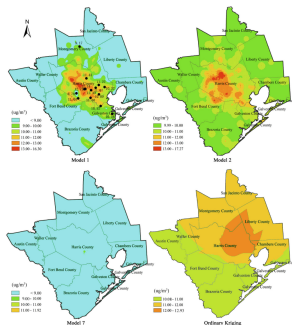
Real-world aim

We want to predict *distribution* and *uncertainty* at unsampled areas using nearby observations.

Solution is **Kriging**:

- **Method definition:** Kriging (noiseless Gaussian Process—GP regression) uses *spatial correlation* to interpolate values at unsampled locations *and* quantify prediction uncertainty via posterior variances.

One real-world application of Kriging is *urban air quality mapping*, Zhai et al. (2017):



Given $PM_{2.5}$ readings from a sparse network of sensors, use Kriging to interpolate pollution levels at unsampled locations across the city, producing both a smooth concentration map *and* pointwise uncertainty for exposure assessment.

Challenge in Standard Kriging

Scalability: Rasmussen and Williams (2006) proved $\mathcal{O}(n^3)$ time and $\mathcal{O}(n^2)$ memory required, where n is the number of observations.

- Solution is Sparse Variational Gaussian Process (SVGP), which provides faster running speed via inducing points; time drops to $\mathcal{O}(nm^2 + m^3)$, m = number of inducing points and $m \ll n$ (Titsias, 2009).
- SVGP approximates complex posteriors by turning inference into an optimization problem within the Bayesian framework by optimizing Evidence Lower Bound (ELBO) (Titsias, 2009).

SVGP Challenge

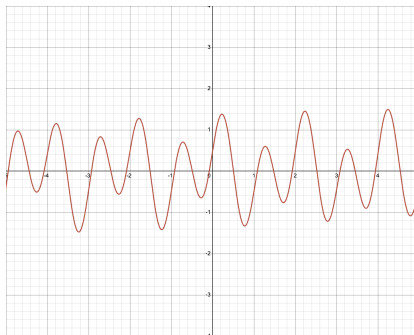
Uncertainty calibration: narrower or wider uncertainty bands when assumptions are off (Titsias, 2009).

In this project, we aim to solve the *undercoverage issue* of SVGP via tempered variational Bayes (TVB) calibration (Yang et al., 2020; Syring and Martin, 2019) while maintaining the computational advantage of SVGP.

- 1 Data Simulation
- 2 Investigate undercoverage with baseline approach
- 3 Demonstrate advantage of proposed method
- 4 TVB calibration methods and results

True Function

Figure: True function: $f_0(x) = \sin(2\pi x) + 0.5 \cos(3x)$



Analytic, bounded, and exhibits multiple characteristic lengthscales, and thus is informative for coverage assessment.

Key Data Definitions

All data are **synthetic**.

- x_{train} : training data for model update. **500** training inputs sampled i.i.d. from a uniform distribution on the range $[-2.5, 2.5]$.
- x_{test} : testing data for point-wise coverage. **50** points uniformly drawn from $[-2.5, 2.5]$.
- All inputs (training and testing) are linearly rescaled so that the raw domain maps symmetrically to $[-1, 1]$.
- The observation generation is noiseless, $y_i = f_0(x_i)$.

Key training configurations:

- R : replication time, 100.
- Epochs / batch size: 1500 / 500
- m : inducing points (SVGP): 50, fixed, standardized

Model and Assumptions

Latent function GP prior:

$$f \sim GP(0, k_\theta(\cdot, \cdot)), k_\theta(x, x') = \sigma_f^2 \exp\left(-\frac{(x-x')^2}{2\ell^2}\right)$$

Sparse variational GP (SVGP):

$$Z = \{z_j\}_{j=1}^m, u = f(Z), q(u) = \mathcal{N}(m, S)$$

Symbol	Meaning
x, x'	1D input locations
$f(x)$	latent GP value at x
$f_0(x)$	data-generating function
y_i	observed response at x_i
k_θ	RBF kernel (squared-exponential)
σ_f^2	prior variance of f

Symbol	Meaning
ℓ	length-scale
$Z = \{z_j\}_{j=1}^m$	inducing input locations
$u = f(Z)$	values at inducing locations
$q(u) = \mathcal{N}(m, S)$	variational posterior of u
<i>Assumptions</i>	smooth f ; noise-free; i.i.d. replications

Posterior distribution & Estimation

Aim to calculate posterior over $f^* = f(X^*)$ under SVGP, Titsias (2009):

$$q(f^*) = \int q(f^*, u) du = \int p(f^* | u) q(u) du.$$

where

- $q(f^*) = \mathcal{N}(\mu^*, \Sigma^*),$
- $\mu^* = K_{X^*Z} K_{ZZ}^{-1} m,$
- $\Sigma^* = K_{X^*X^*} - K_{X^*Z} \left(K_{ZZ}^{-1} - K_{ZZ}^{-1} S K_{ZZ}^{-1} \right) K_{ZX^*}.$

Compute via maximizing **Standard Evidence Lower Bound(ELBO)**:

$$p(f | y) \propto p(f) p(y | f) \iff \mathcal{L} = \mathbb{E}_{q(f)}[\log p(y | f)] - \text{KL}(q(u) \| p(u))$$

Equivalent to minimizing the Kullback–Leibler (KL) divergence between the variational posterior and the true posterior, so the approximate *latent posterior converges* to the true one.

Credible Bands & Coverage Check

Given approximate latent posterior $q(f^*)$, calculate pointwise 95% credible band for each test point x_j :

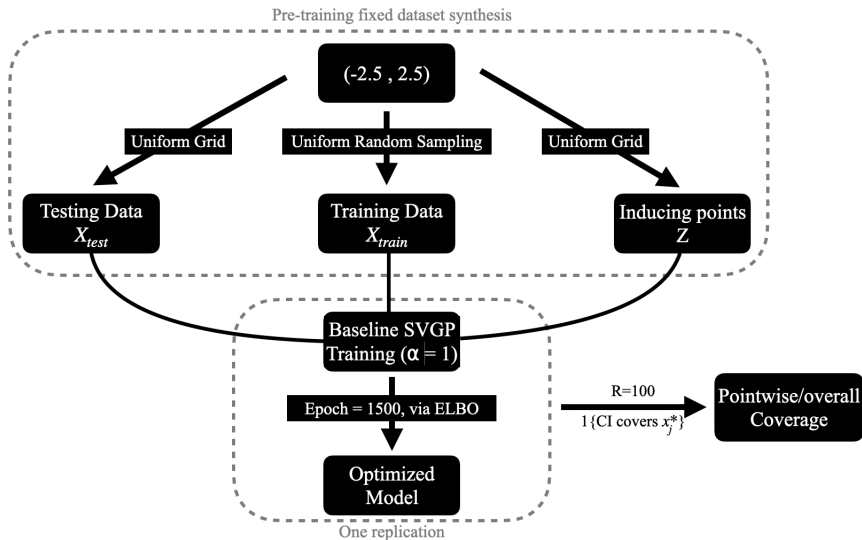
$$\text{CI}_{0.95}(x_j) = \mu_j^* \pm 1.96 \sqrt{\Sigma_{jj}^*}.$$

For each replication r and grid index j :

$$\mathbf{1}\left\{f_0(x_j) \in \text{CI}_{0.95}^{(r)}(x_j)\right\}$$

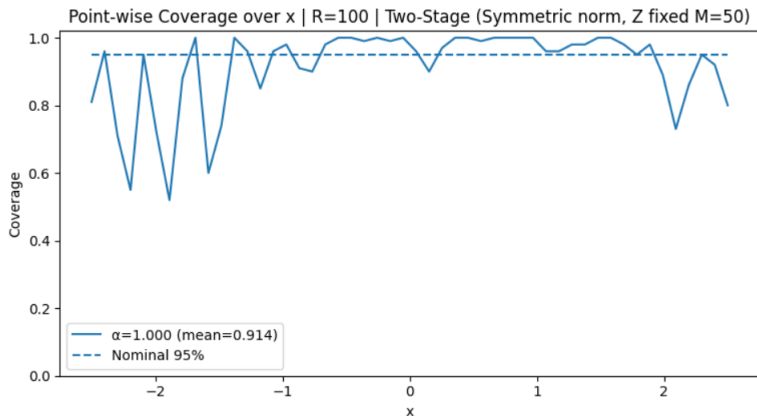
- **Pointwise coverage:** Average indicators for every test point x_j^* ;
- **Overall latent- f coverage:** Average all indicators across R replications.

Replication Protocol



Baseline Coverage Check

After confirming ELBO converges, we evaluate point-wise coverage for $\alpha = 1$ using generated indicators.



This result validates **undercoverage** at baseline level ($\alpha = 1$).

Proposed Method: α -power ELBO

We still need to estimate latent distribution $q(f^*)$ when calibrating the coverage. Thus, we proposed change to the **ELBO equation**:
 α -tempered (power) ELBO

$$p_{\alpha}(f|y) \propto p(f)p(y|f)^{\alpha} \iff \mathcal{L}_{\alpha} = \alpha \mathbb{E}_{q(f)}[\log p(y|f)] - \text{KL}(q(u) \| p(u)).$$

- $0 < \alpha \leq 1$, where $\alpha = 1$ is the standard SVGP model.
- Add **α power** to the likelihood $\iff$ **factor α** to the first term in the ELBO.
- $\alpha \downarrow$ down-weights the likelihood, typically *inflating posterior spread* (better coverage).
- In the SVGP code, α -tempering is implemented simply by setting $\beta = 1/\alpha$ in the MLL function (Gardner et al., 2018).

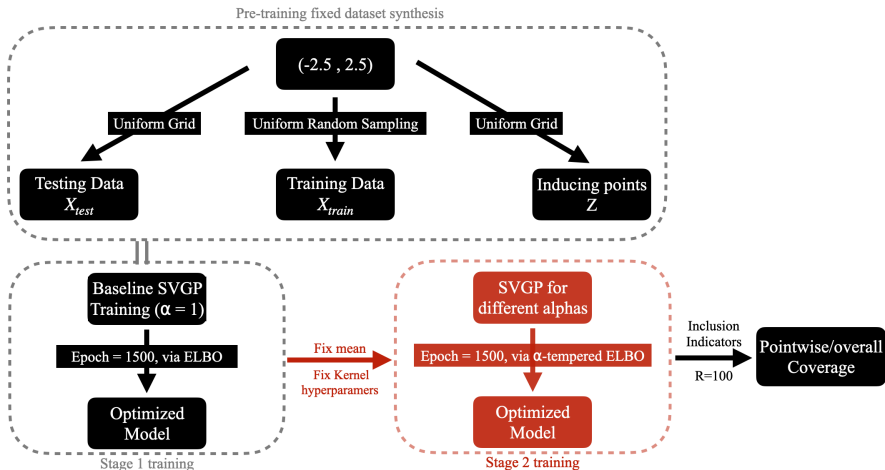
Two-stage Procedure

All configurations and models are identical to the baseline method, including:

- Latent function prior
- Data-generating model
- SVGP variables
- Posterior calculation
- Coverage check procedure

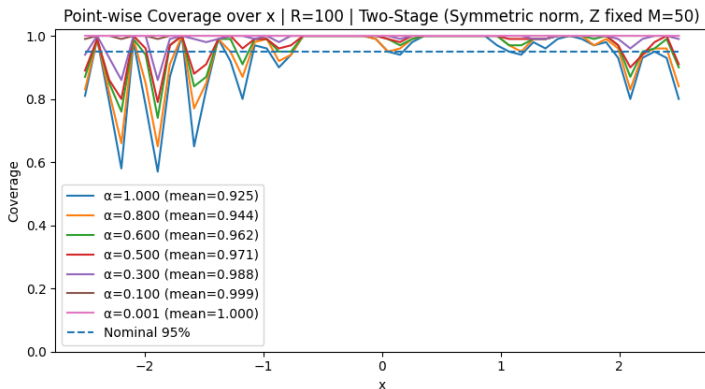
When α is tempered down, the model loses efficiency and accuracy (Pitas and Arbel, 2023). To avoid loss of accuracy, we introduce a **two-stage procedure** that *fixes the mean and kernel hyperparameters learned from baseline $\alpha = 1$* , updating only the latent posterior covariance used for the coverage check.

Two-stage Protocol



Coverage Check with Different α

We evaluate point-wise coverage for multiple alpha values.



As α decreases, **coverage increases monotonically** on every test point.

TVB calibration: big picture

Goal

Calibrate credible bands of latent posterior $q(f^*)$ by grid searching $\mathcal{A}$ that provides near nominal coverage, assuming we **don't** have access to true f_0 .

Idea: Construct a *two-split bootstrap* procedure, and evaluate *pseudo-truth* via a well-fitted SVGP, then study empirical coverage under α -tempered SVGP.

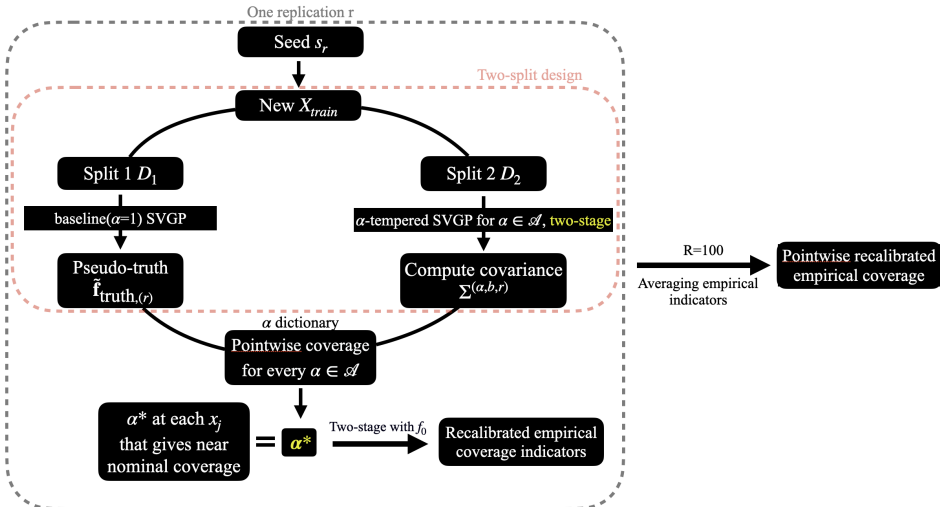
Expected Results

An α -dictionary that allows us to keep track of the optimal α^* for each x_j and in each calibration replication r . Then plug α^* back into the two-stage SVGP pipeline for a *recalibration* experiment to make sure the calibration improves pointwise coverages.

We perform a grid search over $\alpha \in \mathcal{A}$, where $\mathcal{A}$ consists of 100 equally spaced values in $[0.01, 1]$.

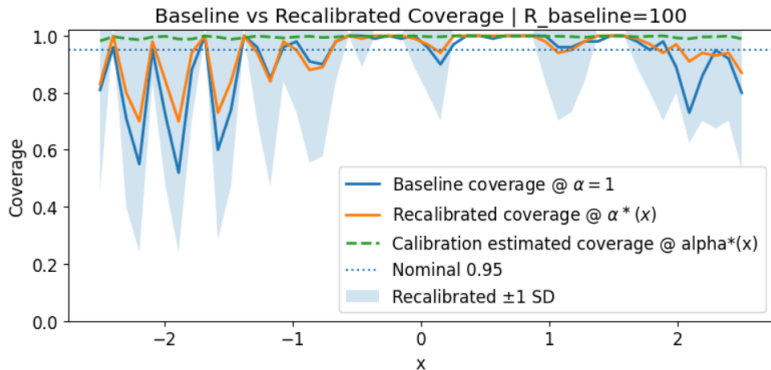
Two-split Procedure

RICE



Recalibration Result

The recalibrated empirical coverage through Two-split protocol shows visible improvement in coverage at most points.



However, at locations where the calibration SD is large (i.e., coverage varies substantially across calibration replications), the recalibrated coverage can still fall below the nominal target.

Conclusion

The results validate under-coverage in the standard SVGP model and also show evidence of monotonic coverage improvements while decreasing power α . TVB calibration indicates merely changing α values can improve empirical coverage while having little effect on prediction accuracy.

Next steps

- Double-check the TVB results, focusing on poor replications
- Calibration for over-covering points
- Current methods is on 1D, next steps could be analyzing 2D or even higher dimensional situations.

References

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Thank you!

Questions welcome.

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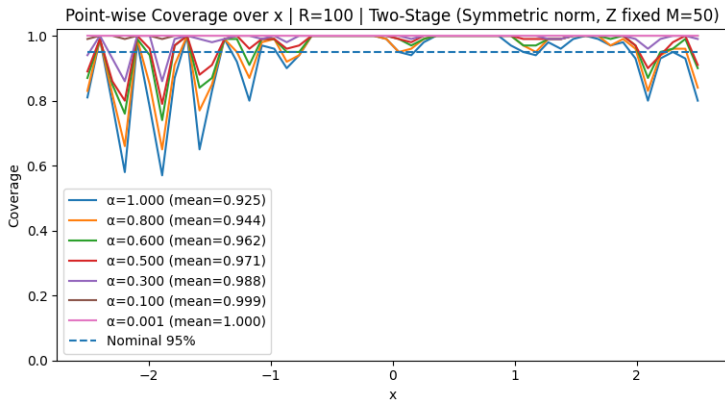
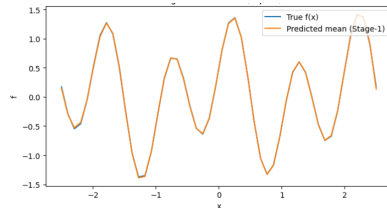
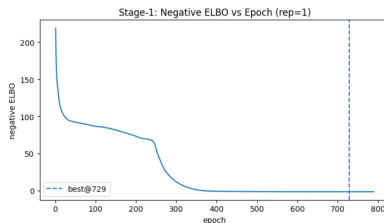


Table: Simulation and training configuration

Quantity	Value
Training size n_{train}	500
Test size n_{test}	50 (uniform grid on $[-2.5, 2.5]$)
Replications R	100
Truth $f_0(x)$	$\sin(2\pi x) + 0.5 \cos(3x)$
Training design	Uniform grid on $[-2.5, 2.5]$
Evaluation target	Latent $f(X^*)$ (noise-free)
SVGP inducing points m	50 (fixed locations)
Epochs / batch size	1500 / 500
Optimizer / LR	Adam / 5×10^{-2}
Kernel	RBF with scale kernel (learned)
Likelihood	Gaussian Likelihood (noise parameter fixed)
Precision / jitter	FP64, jitter 5×10^{-4}
Software	PyTorch 2.5.1, GPyTorch 1.14 (SVGP implementation)

Check the Model – Baseline

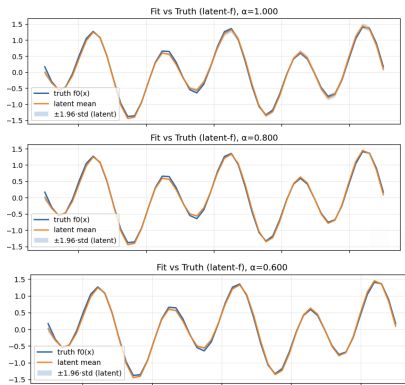
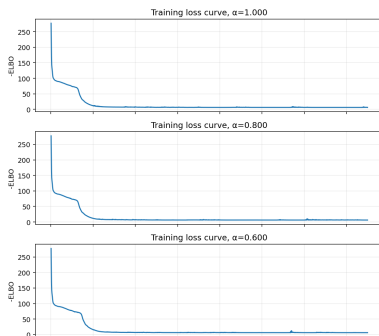
We report negative ELBO loss curve to see if it **converges to a local minimum**; and we compare fit versus true function plot.



These results indicate that the optimization is ideal.

Check the model – Proposed

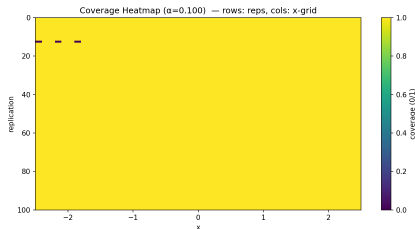
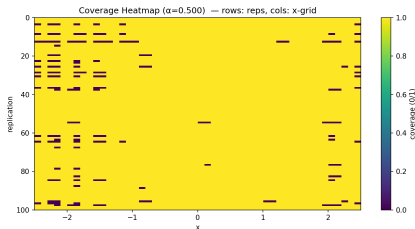
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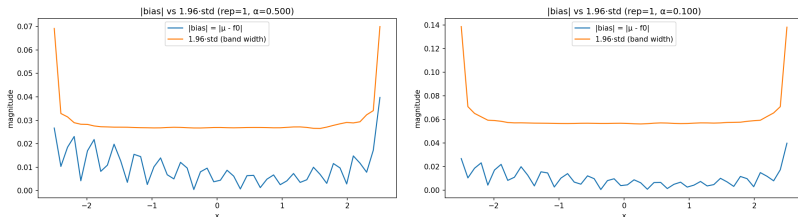
Where Undercoverages Happens

Is it true that undercoverages fall short near boundaries, like the coverage plot?



Two heatmaps visualize indicators for $\alpha \in \{0.5, 0.1\}$ and replications. **Consolidate the "edge effect."**

Why Near Edges



- In the middle of the domain, $|\text{bias}(x)|$ is tiny compared to $1.96\sigma(x)$, so intervals are effectively centered on the truth and coverage stays close to 95%.
- Near the boundaries, predictions are extrapolations: $|\text{bias}(x)|$ becomes much larger relative to $1.96\sigma(x)$, while $\sigma(x)$ does not increase much.
- Because the bands are slightly shifted away from the truth but not correspondingly wider, misses are more frequent at the edges, which shows up as coverage shortfalls near the boundaries.

α -dictionary

[illegible]

Recalibration potential problems

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```

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2.5,0.38,1.0,0.97

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x,alpha_star,cov_indicator,cov_at_alpha_star_calib
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